Phase-1 Corpus Audit and Retraining Plan

Current statistics, problems found, and the methods that address them

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11 October 2026

Outline

1. Current statistics
2. Problems identified
3. Methods
4. Experiment plan and success criteria

Phase-1 pretraining corpus as built today

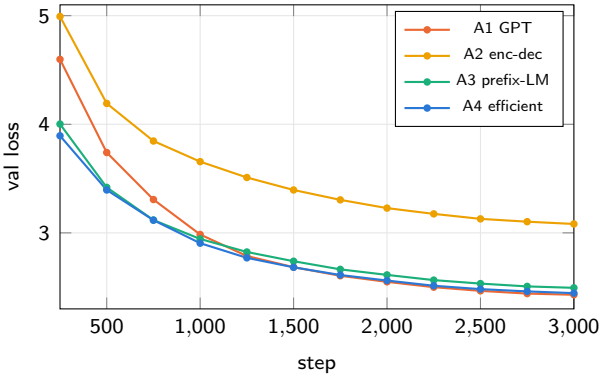
Source	Train docs	Train words	Tokens	Tok/wo
Acts	2,425	14.8M	18.7M	1.
Supreme Court	12,163	44.4M	53.8M	1.
High Courts	120,529	51.9M	66.2M	1.
Tribunals	20,041	37.0M	48.8M	1.
Total	**155,158**	**148.2M**	**187.5M**	1.

Validation: 3,764 documents, 5.0M tokens.
Mix achieved: Acts 10%, SC 28.7%, HC 35.3%, Tribunals 26%.

Pipeline (`legal_lm/data/corpus.py`)

- Stream each source file from the start.
- Keep a doc with probability 0.5.
- `clean_text`: replace NUL, collapse whitespace.
- Drop docs under 50 words.
- 1% of docs to validation, rest to train.
- Stop when the word budget is full.

Phase-1 models: validation loss was still falling



Model	Params	Val ppl
A1 GPT	82.1M	**11.36**
A2 enc-dec	89.4M	21.79
A3 prefix-LM	81.3M	12.11
A4 efficient	87.5M	11.53

- 3000 steps, 196.6M tokens, about 0.75 GPU-hours for A4.
- Curves flatten because the cosine LR ends, not because the models converged.
- Mean GPU utilisation during A4 training: 28%.

Phase 2 (Legal NER) on top of these models

Dev strict macro-F1, mean ± std over 3 dev splits

Run	No post-proc.	Best post-proc.
E2 (A4 + LoRMA)	0.488 ± .007	**0.518 ± .016**
E2 inv-sqrt, r32	**0.497 ± .012**	0.508 ± .013
A4 + LoRA	0.490 ± .018	0.517 ± .043

- Tokenisation is not the bottleneck: oracle macro-F1 is 0.9945.
- Learning curve: 15 docs 0.339, 30 docs 0.448, 60 docs 0.49.

Per-label F1 (A4 + LoRA, dev split 0)

Local pattern		Role in document	
DATE	0.98	APP	0.10
STAT	0.84	RESP	0.08
COURT	0.80	AUTH	0.23
CASENO	0.64	WIT	0.25

Span-only F1 about 0.60 against strict 0.50: about 10 points are lost to the right span with the wrong label.

Problems at a glance

	Problem	Evidence	Severity
A	Under-trained base model	187.5M of >4.5B tokens; about 9× below 20 tokens/parameter; val loss still falling	High
B	Validation leaks into train	29 val docs are exact train copies; ≥52/500 val docs share >50% of 13-grams with train	High
C	No deduplication	1,090 exact duplicates; 1,176 extra copies of the same 300-char prefix	Medium
D	OCR word stutter	566 docs, about 2.4M words (1.6% of the corpus)	Medium
E	Encoding and markup noise	About 70 docs with U+FFFD; RTF control words; 513 Acts docs with long dot/dash runs	Low
F	Short template orders dominate HC	32% of HC docs under 200 words; HC median 291 words	Mixed
G	Token budget missed	Estimated 1.35 tokens/word, measured 1.27	Low
H	Head-of-file sampling	Each file is read from the start until its budget fills	Unverified
I	Long-range NER labels	APP/RESP depend on a header thousands of tokens away; window 512, A4 attention 256	High (Phase 2)

A The base model saw too little text

- The corpus offers more than 4.5B tokens; we used 187.5M (about 4%).
- Rule of thumb of about 20 tokens per parameter gives about 1.7B tokens for an 87M model.
- Val loss of A4 dropped 0.037 over the last 500 steps while the LR was already near its minimum.
- The 200M figure was the course starter value, not a tuned choice.

Consequence

Every later phase inherits a weak backbone. In Phase 2 no adaptation method (full, partial, LoRA, LoRMA, adapters, KD) pushed past about 0.50 macro-F1 without post-processing.

B C Leakage and duplicates

Validation leakage

- The split is drawn per document *before* any duplicate check.
- 29 val documents also appear verbatim in train.
- 52 of 500 sampled val docs share more than half their 13-grams with only 30k of the 155k train docs: a lower bound.
- Phase-1 perplexities are therefore optimistic and checkpoint selection is biased.

Duplicates inside train

- 1,090 exact duplicates (0.7%).
- 1,176 extra copies share the first 300 characters.
- Most common prefix appears 28 times: *"GPN-J-332-ARHCB-8_94-13,10,500-ALA4. SPL.H.C.A.S.C.D. 79e. FARAD FARAD FARAD CONTINUATION SHEET NO. . . . "*

D E Noisy text that survives cleaning

D OCR stutter (Bombay HC scans, 566 docs)

"IN IN THE HIGH COURT OF JUDICAUTRE AT BOMBAY ... CIVIL CIVIL WRIT WRIT WRIT PETITION NO. 7880 OF 2004 Madanlal Veerchand Parekh Parekh Parekh Petitioner Petitioner Petitioner ... CORAM CORAM CORAM"

Teaches repetition, and corrupts exactly the party, counsel and coram region that NER relies on.

E RTF and encoding residue (tribunals)

```
\tab ( L.K. JOSHI )\par
\pard\nowidctlpar\sl480 ... \lquote
sd\rquote\f1\par } ?
```

About 70 docs contain the U+FFFD replacement character.

E Blank-form filler (Acts)

"Signature ..... Date ....."

513 Acts docs (21%) contain runs of 20+ dots, dashes or underscores.

F G H Mixing and sampling

F Template orders

Source	Median	p10
Acts	2,535	410
SC	2,289	432
HC	291	95
Trib.	971	231

Words per document. 38,426 HC docs are under 200 words, mostly one-page orders with the same header template.

G Budget estimate

Budgets are counted in words with 1.35 tokens/word. The measured value is 1.21–1.32, so training got 187.5M instead of 200M tokens.

H Head-of-file sampling

Each file is streamed from its first record and reading stops once the budget fills. If files are ordered by court or year, the sample is skewed. *Not yet verified.*

Role labels need document-level context

- Roles are declared once in the header: *"JAGE RAM & ORS. ..Appellants Versus STATE OF HARYANA ..Respondent"*.
- The same names recur thousands of tokens later, where the label must be inferred.
- Phase-2 documents: median about 4,500 words, max about 20,000.
- The tagger sees 512-token windows; A4 attention reaches 256 tokens.
- Evidence: label propagation across the document was the only reliable gain in the sweep ($+0.02$ to $+0.03$).

Also: annotation ceiling

In the first training document the header party "JAGE RAM & ORS." is not tagged APP, while a later mention is. Some APP/RESP errors are against incomplete gold, which caps achievable F1.

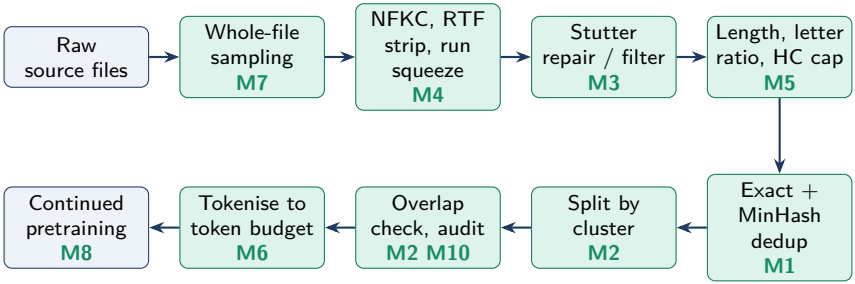
Methods we will apply

	Method	What it does
M1	Exact and near-duplicate removal	Hash of normalised text, then MinHash-LSH on word shingles; keep one doc per cluster
M2	Leak-free split and overlap check	Split by duplicate cluster after M1; assert zero exact and near-duplicate overlap between val and train
M3	OCR stutter repair and filter	Collapse “w w w” to “w”; drop docs whose stutter ratio stays high
M4	Character and markup normalisation	Unicode NFKC; strip RTF control words; drop docs with U+FFFD; squeeze long dot/dash runs
M5	Quality filters and source caps	Per-source minimum length; letter-ratio filter; cap the share of short HC template orders
M6	Measured token budget	Measure tokens per word per source with the actual tokenizer; budget in tokens
M7	Whole-file random sampling	Sample uniformly across each file instead of reading from its start
M8	Continued pretraining at scale	Resume A4 (and A3) on 1–2B cleaned tokens with a re-warmed LR; raise GPU utilisation
M9	Document-context NER	Header-role propagation used in training as well as inference; longer effective context
M10	Audit as a regression test	Re-run `audit.py` on every new corpus; every problem count must reach its target

Which method tackles which problem

	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10
A Under-trained base						●	●	●		
B Val leakage	●	●								●
C Duplicates	●									●
D OCR stutter			●		●					●
E Encoding/markup noise				●	●					●
F Short template orders	●				●					●
G Token budget missed						●				
H Head-of-file sampling							●			●
I Long-range NER labels								●	●	

New data pipeline



Green steps are new. Today's pipeline only has the grey boxes plus NUL removal, whitespace collapse and a 50-word minimum.

M1–M5 Cleaning methods in detail

M1 M2 Deduplication and split

- Exact: hash of lower-cased, whitespace-normalised text.
- Near: MinHash signatures over word shingles, LSH buckets, Jaccard threshold to be chosen on a labelled sample.
- Assign whole clusters to train or val, so no near-copy straddles the split.
- Check after splitting: 0 exact copies, 0 val docs above the overlap threshold.

M3 OCR stutter

- Repair: replace a word repeated three times in a row with one copy.
- Filter: drop docs where repairs touch a large share of the words.

M4 Characters and markup

- Unicode NFKC normalisation.
- Strip RTF control words (`\par`, `\pard`, `\tab`, `\f1`, ...).
- Drop docs containing U+FFFD.
- Squeeze runs of dots, dashes, underscores to three characters.

M5 Quality and mixing

- Per-source minimum length instead of one global 50 words.
- Drop docs with a low share of letters.
- Cap the token share of short HC orders; keep enough for header patterns.

M6–M10 Scale, sampling, NER and checks

M6 M7 Budget and sampling

- Tokenise a sample per source and use the measured tokens/word.
- Count the budget in tokens at the tokenise step.
- Index record offsets, or use reservoir sampling, to sample evenly across each file.

M8 Continued pretraining

- Resume from `runs/A4_efficient/last.pt` (and A3).
- 1–2B cleaned tokens, LR re-warmed then decayed again.
- Raise throughput from 28% GPU utilisation: larger micro-batch, `torch.compile`, more loader workers.

M9 Document-context NER

- Extract header roles (Appellant / Respondent / Petitioner lines).
- Apply label propagation during training, not only at inference.
- Larger windows where memory allows.

M10 Audit as a regression test

- Same script as this audit, run on each new corpus.
- Fails the build if any count in the targets table is above its target.

Experiment plan: separate the effect of cleaning from scale

Run	Change	Tokens	Isolates
R0	Current A4 checkpoint, re-evaluated on the new leak-free val set	187.5M	Baseline
R1	A4 from scratch on cleaned data (**M1–M7**), same budget	200M	Effect of cleaning
R2	R1 continued on more cleaned data (**M8**)	1–2B	Effect of scale
R3	R2 plus Phase-2 training texts in the mix (domain-adaptive)	R2 + small	Effect of in-domain text
N1	Phase-2 E2 recipe on R1 / R2 / R3 backbones	–	Downstream gain
N2	N1 best backbone + document-context NER (**M9**)	–	Effect of **I** fix

Fair comparison

All runs, including R0, are scored on the same new leak-free validation set. Phase-2 numbers are mean $\pm$ std over dev splits 0, 1, 2, with and without post-processing.

Success criteria

Corpus audit targets (M10)

Check	Today	Target
Val docs exactly in train	29	0
Val docs >50% 13-gram overlap	≥52/500	0
Exact duplicate train docs	1,090	0
Docs with heavy OCR stutter	566	0
Docs with U+FFFD	about 70	0
Train tokens vs budget	93.7%	≈100%

Model targets

- R1 vs R0: lower val perplexity on the leak-free val set at equal tokens.
- R2 vs R1: val loss keeps falling with more tokens.
- N1 vs current E2: higher Phase-2 macro-F1 with no post-processing (today 0.488 ± .007).
- N2: APP and RESP F1 well above today's 0.10 / 0.08.

Summary

Problems

- **A** Base model trained on about 4% of the corpus.
- **B C** Val leaks into train; duplicates kept.
- **D E** OCR stutter, RTF and encoding noise.
- **F G H** Skewed mix, missed budget, head-of-file sampling.
- **I** NER role labels need document context.

Methods

- **M1 M2** Deduplicate, then split by cluster.
- **M3 M4 M5** Repair and filter noisy text.
- **M6 M7** Measured budget, whole-file sampling.
- **M8** Continue pretraining on 1–2B clean tokens.
- **M9** Document-context NER.
- **M10** Audit every new corpus.